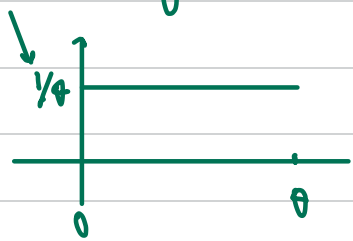
MAXIMUM LIKELIHOOD ESTIMATOR

- $x_1, \dots, x_n$ i.i.d. sample, x_i comes from X_i has pdf $P(x;\theta)$
- PROBLEM: what is θ? The MLE method is a very general approach to learn θ.

EXAMPLE: $x_1, \dots, x_n$ i.i.d. $x_i \sim U(0,\theta)$ $P(x;\theta) = \frac{1}{\theta}$ if $0 \le x \le \theta$

- IDEA: Consider the pdf of the sample given θ.



$$P_{X_1,\dots,X_n}(x_1,\dots,x_n;\theta) = \prod_{i=1}^{n} P(x_i;\theta) = \ell(\theta)$$

$$= \prod_{i=1}^{n} \frac{1}{\theta} \quad \text{provided all } x_i < \theta.$$

The higher $\ell(\theta)$, the "more likely" it is to observe the sample $x_1, \dots, x_n$ given θ. If $\ell(\theta_1) < \ell(\theta_2)$ then we tend to prefer θ_2.

EXAMPLE: $\ell(\theta) = \frac{1}{\theta^n}$ provided all $x_i \le \theta$ otherwise $\ell(\theta) = 0$

assume $(x_1, x_2, x_3) = (0.4, 1, 2)$ $\ell(3) = \frac{1}{3^3}$ $\ell(2) = \frac{1}{2^3}$

$\ell(1) = 0$. We prefer $\theta = 2$ between 1, 2, 3

- IDEA: Find $\hat{\theta}$ that maximises $\ell(\theta)$. We call this $\hat{\theta}$ MAXIMUL LIKELIHOOD ESTIMATOR of θ.

EXAMPLE: $\ell(\theta) = \frac{1}{\theta^n}$ provided $x_i \le \theta$. Here $\underset{\text{MLE}}{\hat{\theta}} = \max_i x_i$

If $\theta < \max_i x_i \Rightarrow \ell(\theta) = 0$

If $\theta \ge \max_i x_i \Rightarrow \ell(\theta) = 1/\theta^n$ decreasing, so we want θ as low as possible.

GENERAL MAXIMUM LIKELIHOOD METHOD

Assume $x_1, \dots, x_n$ i.i.d. from $P(x_i; \theta)$.

To find the maximum likelihood estimator of θ:

STEP 1 Compute $\ell(\theta) = \prod_{i=1}^{n} P(x_i; \theta)$ (iff $x_1, \dots, x_n$ i.i.d.)

optionally set $L(\theta) = -\log(\ell(\theta))$

STEP 2 Maximise $\ell(\theta)$ and set $\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\ \ell(\theta)$

optionally minimise $L(\theta)$

EXERCISE 1

(a) Find $P(x;\theta)$ which is the derivative of the cdf

$$P(x;\theta) = \theta x^{\theta-1} \quad \text{for } x \in (0,1), \quad 0 \text{ otherwise}$$

(b) If $x_1, \dots, x_n$ close to 1 I expect θ to be higher than one tbh!

(c) $\ell(\theta) = \prod_{i=1}^{n} \theta x_i^{\theta-1}$ likelihood of θ given $x_1, \dots, x_n$

STEP 1 computing $\ell(\theta)$

$$= \theta^n \prod_{i=1}^{n} x_i^{\theta-1}$$

← too annoying to calculate the derivative of this! Instead:

$$L(\theta) = -\log \ell(\theta) = -\log(\theta^n) - \log\left(\prod x_i^{\theta-1}\right)$$

$$= -n\log(\theta) - \sum_i (\theta-1)\log x_i$$

$$= \theta \sum \log x_i - \sum \log x_i$$

STEP 2 : minimise $L(\theta)$

$$L'(\theta) = \frac{-n}{\theta} - \sum \log x_i = 0$$

$$\hat{\theta}_{MLE} = \frac{-n}{\sum \log x_i}$$ MAXIMUM LIKELIHOOD ESTIMATOR OF θ

(check the second derivative if you like)

EXERCISE 3 (sketch solution)

$x_1, \dots, x_n$ i.i.d. $\quad x_i \sim \text{Poisson}(\mu)$

$$P(x;\mu) = e^{-\mu} \frac{\mu^x}{x!}$$

So far we used as estimate for μ the sample mean $\bar{x}$

Let's calculate MLE of μ instead

STEP 1

$$\mathcal{L}(\mu) = \prod_i e^{-\mu} \frac{\mu^{x_i}}{x_i!}$$

$$L(\mu) = -\log(\mathcal{L}(\mu)) = -\sum \log\left(e^{-\mu} \frac{\mu^{x_i}}{x_i!}\right)$$

$$= -\sum \underbrace{\log(e^{-\mu})}_{-\mu} - \sum \log(\mu^{x_i}) + \sum \log x_i!$$

$$= n\mu - \log(\mu) \sum x_i + \sum \log x_i!$$

STEP 2

$$L'(\mu) = n - \frac{\sum x_i}{\mu} = 0 \qquad \underset{\text{MLE}}{\hat{\mu}} = \frac{\sum x_i}{n} = \bar{x}$$